

OPHIRA

Reflectometry-fitting toolkit for porous silicon and thin films

D1 — User operating manual

Detailed description of every window, button, slider and interface control. Includes the recommended operating workflows (quick / rigorous / save-share) and the formats of the files produced/read by the program. It does not cover the physics and mathematics of the forward model, which are documented in the physics & math reference.

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Related documents	D2 (physics & math reference), D3 (operating procedures)

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1 Introduction

OPHIRA is a PyQt application that reconstructs the complex optical indices $n(\lambda)$ and $k(\lambda)$ of thin films from reflectance spectra taken at several angles of incidence. It was developed mainly for porous silicon (PSi), a material where standard dielectric models (Cauchy, Tauc-Lorentz, EMA) fail because n and k depend on the morphology (pore shape, distribution, connectivity) and not on porosity alone.

The methodological strategy is **data-first, physics-as-validation**: a free-node fit with no imposed dielectric model, subject to the physical constraints of smoothness, monotonicity and optical-thickness anchoring; followed by verification through Kramers-Kronig (KK) and geometric thickness triangulation. The software treats separately the small-angle fit (8° , 20°) and the highly oblique $40^\circ/60^\circ$ fit with angular spread and birefringence (rendered as $\Delta n_{\text{eff}} = n_{\text{ang}} - n_{8^\circ}$; the model is non-polarized throughout).

1.1 High-level architecture

The application consists of four main windows, each with its own purpose:

- **Main Window** — the main window where the user adjusts the sliders, launches the fits and views the spectrum, residual, $n(\lambda)$ and $k(\lambda)$.
- **KK Window** — Kramers-Kronig consistency analysis at 8° with an RMS/ σ metric and transfer of the UV nodes.
- **Peak Analysis** — reflectance-maxima analysis for an independent thickness estimate via Bragg/Snell.

All the state (parameters, sliders, per-angle data, materials, `ext_override`) is centralized in a single `AppState` object shared by reference across the windows and the fit engine.

1.2 Requirements

- Python 3.10 or later.
- PyQt6 (preferred) or PyQt5 as automatic fallback.
- numpy, scipy, matplotlib.
- The optical constants of the substrates and of the known films live in `materials/substrates/` and `materials/films/` (columns: `wl_nm`, `n`, `k`; the leading # lines carry the source of the data). The two folders **are** the menus: see §3.

1.3 Startup

```
python main.py
```

At startup the app clears the material cache (see `physics._reload_material_cache`) and opens the **Main Window** with an empty spectrum. The user loads the data through the **LOAD** buttons or the four per-angle buttons at the top right. The other windows open on demand from the analysis button row.

2 Main Window — layout

The Main Window is divided into five functional zones:

- **Left panel** (sliders, scrollable) — 15 sample/instrument parameter sliders + 14 $n(\lambda)$ sliders + 14 $k(\lambda)$ sliders. Above each n/k column there is a small (padlock) button for bulk locking.
- **Top-right panel** — four angle-selection buttons (**8°**, **20°**, **40°**, **60°**) each with a (folder) button next to it to load the corresponding spectrum, followed by the two drop-down menus **Substrate** and **Film**.
- **Central plot** — a matplotlib canvas with two axes: on top the experimental and fitted reflectance $R(\lambda)$ + residual, at the bottom $n(\lambda)$ and $k(\lambda)$ with $\pm\sigma$ uncertainty bands.
- **Bottom button bar** — three rows of operating buttons grouped by family (file I/O, fit, analysis/toggles).
- **Status bar** — bottom line showing messages about loading, saving, fit in progress, errors.

2.1 Left panel — Parameters sliders

The **Parameters** column (purple header) groups 15 sliders for scalar sample and instrument parameters. The order and ranges are:

The **Default** column below lists each slider's *nominal* value. At startup and after RESET the program does **not** load these: it loads a built-in demonstration PSi fit ($d \approx 1200$ nm), so on launch the sliders show that fit's values, not the nominal ones. The nominal column is a neutral reference for the meaning and sign of each slider.

Slider	Range	Default	Meaning
d	10 – 6000	1000	Film thickness in nm. Initial estimate; refined by the fit.
Scatt.	0 – 300	5	Overall scattering. Reduces fringe amplitude without shifting their position.
Inhom. %	0 – 10	0	Lateral inhomogeneity (% variation of d over the footprint). Reduces fringe amplitude at large angles; degenerate with Scatt.
Offs	0 – 20	1	Additive offset of $R(\lambda)$ (compensates instrumental dark/baseline).
IR Jump	0.2 – 2.0	1.0	Coefficient k_m of the IR correction (multiplicative jump).
IR Exp	–4.0 – 4.0	0.0	Exponent k_s of the IR correction (spectral slope).
IR Dmp	0.0 – 3.0	0.0	Damping k_b of the IR correction (curvature in the rolloff).
W 300	280 – 320	300	Position (nm) of the 3rd UV base node — free for PSi fits.
W 335	315 – 355	335	Position of the 4th base node — free for PSi fits.
W 375	350 – 400	375	Position of the 5th base node — free for PSi fits.
W 400	390 – 450	410	Position of the 6th base node — free for PSi fits.
W 1100	950 – 1200	1100	Position of the 11th base node (IR) — free for PSi fits.
Fr contrast	0.0 – 3.0	1.0	Fringe-contrast envelope: baseline amplitude of the fringe contrast.
Fr ctr vtx	–2 – 2	0.0	Fringe-contrast envelope: vertex wavelength λ_v of the contrast-vs- λ parabola (shown in nm; $\lambda_v = 1500 + 1000 \cdot x$).

Slider	Range	Default	Meaning
Fr ctr curv	-2 – 2	0.0	Fringe-contrast envelope: curvature of the contrast vs λ around the vertex.

The five sliders **W 300**, **W 335**, **W 375**, **W 400**, **W 1100** shift the λ position of base nodes of the n,k model. They are free only for the PSi film (film_mat = 0); for known films (SiO₂, TiO₂, ...) they stay fixed at the tabulated values.

2.2 Left panel — $n(\lambda)$ and $k(\lambda)$ sliders

The two columns (green and purple) each contain **14 sliders**, one per base node, labelled in nm with the node's wavelength. The 14 nodes are fixed at the positions:

240, 270, 300, 335, 375, 430, 470, 520, 650, 900, 1100, 1583, 2067, 2550 nm

The positions 300, 335, 375, 400, 1100 are modulated by the corresponding W sliders (see above). The numeric value shown next to each slider is already n (for the green column) or k (for the purple one); internally the app uses a non-linear mapping (n_{map} , k_{map}) to give high resolution in the regimes of interest.

2.2.1 Locking individual sliders

Each slider has a small padlock on the right: when clicked, it fixes that parameter's value during the next fit. The lock state persists in state. lock_left / lock_n / lock_k / lock_extra.

2.2.2 Bulk lock (padlock) above the n/k columns

The (padlock) button above the $n(\lambda)$ column locks/unlocks all 14 n sliders at once (toggle). Same for the $k(\lambda)$ column. Useful for fixing an entire curve during a phase of the KK workflow or for comparisons.

3 Main Window — right panel and selectors

3.1 Angle buttons

At the top right there are four rows, each with an **angle** button (8°, 20°, 40°, 60°) followed by the per-angle (**folder**) load button:

- **Angle button (8°/20°/40°/60°)**: selects the active angle. When an angle has no loaded data the button is grey and disabled. Once data are loaded it turns light green; the one currently selected is dark green with white text. Clicking it changes `state.current_ang`, restores the parameters saved for that angle (if any from the previous fit) and updates the plot.
- **Per-angle load (folder icon)**: opens a dialog to choose the spectrum file to load. Unlike the main **LOAD**, it does not ask for the angle (it is already fixed by which (folder) was clicked). Suitable for quick workflows when loading files one by one.

3.2 Substrate combo

Drop-down menu listing the available substrates: one entry per CSV in `materials/substrates/` (**Si** always first), plus **Air**, which needs no file because $n=1$ by definition. The chosen substrate enters the Fresnel calculation of the fit. Changing it also updates `physics._substrate[0]`.

3.3 Film combo

Drop-down menu of the film materials: one entry per CSV in `materials/films/` (**SiO₂**, **SiO₂_thermal**, **TiO₂**, **Si**, **Al₂O₃** as distributed), preceded by **PSi**. **PSi** is not a file: it is the free fit — choosing it leaves the n,k sliders and the W-nodes free (free-node mode). For a known film the n,k sliders are locked to the tabulated values and the W-nodes are fixed.

3.4 Adding a material

Drop a `wl_nm,n,k` CSV into `materials/films/` or `materials/substrates/`: at the next start the file name is the name in the menu. The same material can serve both roles — then it needs a copy in each folder. A file that does not parse is reported on the console and left out of the menu, so a material can never be selected without its data. Sessions and `.par` files store the material **name**, so adding or removing a CSV never re-labels a saved fit.

The combo also includes the entry «— Load user material... —» (disabled for now) as a future placeholder for ad-hoc loading.

If the whole `materials/` folder is missing, the program still runs on a minimum internal database (**Si**, **BK7**, **SiO₂**, **Sapphire**, **CaF₂**, **Air**; films **SiO₂**, **TiO₂**, **Si**, **Al₂O₃**) computed from the published Sellmeier equations — with the warning of §11 for Si.

3.5 FIT / MAXIMA view toggle (per-angle)

Above the right panel, a **View: FIT / View: MAXIMA** button switches how the current angle is presented in the main plot. It is a display-only override (it does not re-fit, it does not alter the data):

- **View: FIT** (default): shows $n(\lambda)$ and $k(\lambda)$ from the fit, with the residual $R_{\text{fit}} - R_{\text{exp}}$.
- **View: MAXIMA**: shows $n(\lambda)$ derived from the *maxima positions* (n_{eff} method, described in D2). The residual is zeroed and a per-angle *disclaimer* appears. At **60°** the k is hidden (at oblique incidence the maxima give only an effective index, not k) and the disclaimer is strong; at the other angles it is milder (« k from fit»).

Prerequisite for View: MAXIMA — having computed peaks / m / d in the Peak Analysis window. If missing, the toggle tries to compute n from maxima on the fly, otherwise it warns.

3.6 «Tools» menu

In the top menu bar, the **Tools** menu holds **Compare with Scout...**: a read-only window that puts this program's $n(\lambda)$, $k(\lambda)$ and $R(\lambda)$ side by side with the ones exported by Scout, for the same sample. It reads files and draws — it never touches the fit or the state.

3.7 «Windows» menu

The **Windows** menu lists the windows opened from here — the d-scan analyses (each with a distinguishing name) and the Scout comparison — brings any of them to the front, and closes them all together with **Close all windows**. Handy when several cascaded d-scan analyses are open at once.

4 Main Window — bottom button bar

The operating buttons are grouped in three horizontal rows at the bottom of the window. The following section covers them one by one, grouped by row.

4.1 Row 1 — file I/O

Button	Where	What it does / when to use it
LOAD	Row 1, blue	Loads a spectrum. Auto-detect of the format: if the file starts with the marker '# PSi Reflectometer manifest' it loads the data of all the angles listed in the manifest without a dialog. Otherwise it asks for the angle and loads a single raw spectrum. It loads only the R_{exp} curves: no parameters, no fit, no nodes. For parameters use LOAD PAR; for everything, LOAD SESS.
SAVE	Row 1, plum	Saves the data and parameters of the currently active angle. Produces <i>name_{ang}deg.par</i> (all parameters + 14 base nodes with σ) and <i>name_{ang}deg_data.csv</i> (R_{exp} , R_{fit} , continuous n , k + optional n_{KK} , k_{KK} and σ_n , σ_k at wl_{exp}). Asks for sample name and folder.
SAVE ALL	Row 1, light plum	Like SAVE but for <i>all</i> the angles with data. The file names include the suffix <i>_all_</i> (e.g. <i>Sample_all_8deg.par</i>). It also produces a manifest file <i>name_all.csv</i> which is the single entry point to reload all the data at once.
LOAD PAR	Row 1, gold	Loads a saved .par file (single angle). Updates sliders, symbolic materials, W-nodes, HOBL parameters. Whatever the file does not contain is left as it is: a material it does not name keeps the one currently selected.
EXIT	Row 3, red	Closes the application.
SAVE SESS	Row 1, powder	Saves the whole session to a single .psi_sess file (binary pickle, not readable in an editor). Includes data, parameters, ext_override, sliders, kk_cache. For internal backup and exact resume.
LOAD SESS	Row 1, powder	Loads a session saved with SAVE SESS. Completely overwrites the current state.
EXPORT FIG	Row 1, amber	Exports the current figure of the main plot in high resolution (PNG/PDF) for publication. Opens a dialog to choose name and folder.

Button	Where	What it does / when to use it
d-SCAN	Row 3, light green	Opens the d-scan analysis from N saved sessions (semi-automatic): you select several .psi_sess files of the same sample, fitted at different thicknesses. It produces the d-scan analysis windows described in the dedicated section below — the $n(\lambda)$ overlay (the most reliable d-indicator: the smoothest, most Cauchy-like curve wins), the ΔR d-discriminants (a corroborating signed estimator), the exp-vs-sim reflectance comparison, and a complementary RMS-residual(d) / KK-residual window (with an abs $\leftrightarrow$ signed KK toggle and an adjustable spectral band). Every figure window has publication-export controls (Save PNG/PDF, an aspect-ratio lock and draggable legends). The figures also auto-save to _paper_figures/. The detailed procedure is in D3.
ABOUT	Row 1, right (purple)	Shows version, build and credits of the application.

No consistency check on spectrum loading. The program does *not* verify that the loaded spectrum matches the angle of the slot it is placed in: it is up to the **user** to make sure the right file is loaded into the right angle. A wrong file in a slot (e.g. the 60° spectrum loaded into the 40° slot) produces a poor fit at that angle *with no warning* — the engine tries to fit those data with the slot's angle model. Always check the loaded files, especially when the names are similar (e.g. ..._40° / ..._60°).

4.2 Row 2 — fit and angle change

Button	Where	What it does / when to use it
FIT	Row 2, green	Runs the fit of the <i>currently active angle</i> . For 8°/20° it runs synchronously in the UI thread (with throttling of the graphic updates); for 40°/60° (HOBL) it runs in a separate thread. Updates sliders, ext_override and R_fit. Computes σ_n/σ_k via the diagonal Hessian and saves them in state.data_angles[ang]['sigma_nk'].
FIT MULTI	Row 2, light blue	Multi-angle fit with consistency constraints. Weights 8° most, decreasing at 20°, 40°, 60°. Three internal stages: stage 1 = fit common to all n,k nodes + instrumental parameters; stage 2 = per-angle refinement; stage 3 (HOBL only) = a multi-start to escape local minima. Ends with a single fit for each angle as refinement.
FIT AUTO	Row 2, pale green	End-to-end workflow. First part: self-consistent loop with the Peak Analysis window to determine the thickness d via Bragg/Snell on the reflectance maxima. Second part: calls FIT MULTI for the global fit, and ends with single fits for a final refinement. Typical starting point for a new sample.
STOP	Row 2, orange	Requests termination of the current fit (sets state.stop = True). Useful if you see that convergence is not progressing or you want to save what has already been reached.
COPY 8°	Row 2, gold	Copies the 8° angle fit (ext_override) into the currently active angle. Requires that 8° has already been fitted. Useful as a <i>seed</i> for the 20°/40°/60° fit, since the n,k of a sample are the same and only the instrumental effects change.
MONO n ON / OFF	Row 2, green/grey (toggle)	Enables/disables all the shape constraints on $n(\lambda)$: monotonic decrease and convexity in the region $\lambda \geq 420$ nm, plus the anti-zigzag term (below 420 nm there is anomalous dispersion of P <i>Si</i> and the constraints are disabled by construction). Typically ON for standard fits.
MONO k ON / OFF	Row 2, green/grey (toggle)	Enables/disables all the shape constraints on $k(\lambda)$: monotonic decrease and convexity in the region $\lambda \geq 420$ nm (the same as n), plus the anti-zigzag term. For P <i>Si</i> , $k(\lambda)$ must decrease monotonically in the visible and vanish in the IR.
d src (combo)	Row 2, light purple	Source of the peaks for computing d : exp (peaks from the raw data, robust default), fit (model peaks, useful at convergence), hybrid (fit peaks as a guide + exp maxima in a ± 30 nm window). Affects both the Peak Analysis window and FIT AUTO, keeping them consistent.
THIN FILM	Row 2, grey/orange (toggle)	Thin-sample mode (d below ~ 700 nm): lowers the accepted λ threshold from 1000 to 600 nm, accepts even a single NIR peak ($m=1$ assumed) and enables the single-angle formula (more stable on thin samples).

4.3 Row 3 — analysis, toggles, auxiliaries

Button	Where	What it does / when to use it
PEAK ANAL.	Row 3, gold	Opens the Peak Analysis Window . See dedicated section.
PRED 8° ON / OFF	Row 3, orange/grey (toggle)	Shows the predicted curve at 8° using the current n, k , overlaid on the spectrum. Useful as a visual diagnostic of transferability between angles.
KK ON / OFF	Row 3, accent/grey (toggle)	Opens the Kramers-Kronig window (only if the active angle is 8° — KK is valid only at near-normal incidence) and shows in the main plot the n_{KK} and $\Delta n = n_{\text{fit}} - n_{\text{KK}}$ curves. At 40°/60° it warns and does not open. Not to be confused with KK→UV below.
KK→UV	Row 3, light grey	Applies the n_{KK} values to the UV nodes (the first ~5) using the last cached KK calculation. Works only after both the fit and the KK calculation have been done. Equivalent to the APPLY KK→NODES button inside the KK window but callable from the main window.
MEAS JUMP	Row 3, amber	Measures the amplitude of the detector jump at ~890 nm directly from the current angle's data, sets it on the <i>IR Jump</i> slider and locks it. Data-anchored: prevents the fit from confusing the instrumental jump with $n/k/\text{scatt}$. The measured value varies by sample/angle.
RESET	Row 3, red	Restores the default first-launch curve — a clean PSi fit at $d \approx 1200$ nm (its n/k , thickness and instrumental parameters), the same curve shown at startup, not flat values. Asks for confirmation.

Next to the row-3 buttons, a **Spread** label shows the width of the angular spread currently in use (HOBL configuration at 40°/60°).

5 Kramers-Kronig window

It opens by clicking **KK** in row 3 of the Main Window (only if the active angle is 8° — KK is physically valid only at near-normal incidence). The window is dedicated to the consistency analysis between $n(\lambda)$ obtained from the fit and $n(\lambda)$ reconstructed via the Hilbert transform from the fitted $k(\lambda)$.

5.1 Layout

Three stacked plots on the left (n_{fit} , n_{KK} and Δn), a controls panel on the right, a scrollable table of the values at the nodes.

5.2 Controls — RMS range

Slider	Range	Default	Meaning
λ_{min} RMS (edit)	200 – 600	420	Minimum wavelength included in the $\text{RMS}(n - n_{\text{KK}})$ calculation. Below λ_{min} (default 420 nm, the anomalous-dispersion boundary) the region is data-limited and excluded from the KK test.
λ_{min} slider	200 – 600	420	Slider version of the field above. The two update each other.
λ_{max}	fixed 2550	2550	Upper end of the Hilbert integral (known boundary artifact at the 2550 node, see note).
Smoothing window	1, odd	1	Savitzky-Golay window (polyorder=3) applied to n_{KK} . 1 = no smoothing; typical values 5–15.

The node at **2550 nm** is subject to a boundary artifact from the truncation of the Hilbert integral. If in the $\text{KK} \rightarrow \text{fit} \rightarrow \text{KK}$ cycles you see it drifting downward, manually lock that node to the value from the previous cycle.

5.3 Controls — Display range

A separate view-range control (min/max fields + sliders, plus a **Full range** button) sets *only* what is shown on the plots — handy to zoom into the same interval another tool (e.g. Scout) displays. It does **not** change the KK calculation, which always runs on the full spectrum: the Hilbert transform is non-local, so truncating the input would corrupt n_{KK} . **SAVE CURVES** exports over the current view range.

5.4 KK self-consistency metrics

At the bottom right the panel shows four metrics updated at every recalculation:

- **$\text{RMS}(n - n_{\text{KK}})$** : root-mean-square deviation over the interval $[\lambda_{\text{min}}, 2550]$. Main consistency indicator.
- **$\max|\Delta n|/n$** : maximum relative error (to identify outlier nodes).
- **σ_{KK} (noise)**: local standard deviation of the raw n_{KK} , computed on a sliding window. Measures the intrinsic noise of the KK calculation independently of the smoothing.
- **$\Delta n/\sigma_{\text{KK}}$** : signal-to-noise of Δn . Values of 2–3 are the noise level (consistency within the limits of the KK).

5.5 Nodes table

Scrollable, shows for each base node: λ , n_{fit} , n_{KK} , Δn , $\sigma_{\text{KK_local}}$, $\Delta n/\sigma$. Highlighted row for the nodes in the range $[\lambda_{\text{min}}, 2550]$.

5.6 Action buttons

Button	Where	What it does / when to use it
RECALCULATE	KK, Actions panel	Reruns the KK calculation from the current n,k (sliders or ext_override) with the chosen smoothing and $\lambda_{\min}$ parameters. To be used at every significant change in the main window.
APPLY KK→NODES	KK, Actions panel	Transfers the n_KK values to the UV nodes (excluding the end nodes). Equivalent to KK→UV in the Main Window. Resets ext_override = None so that the new values are reflected in the sliders.
SAVE CURVES	KK, Actions panel	Saves a single CSV file with the curves shown in the plot, over the current display range: wl, n_fit, n_KK_smooth, n_KK $\pm$ σ_{KK} (band). Lightweight, ideal for replotting in Origin/Excel.
SAVE ALL	KK, Actions panel	Saves 3 complete files: _kk_grid.csv (dense grid with n_KK_raw, n_KK_smooth, k_input_Hilbert, $\sigma_{\text{KK_local}}$, $\sigma_{\text{KK_propagated}}$, Δn), _kk_nodes.csv (base nodes with $\Delta n/\sigma$ and in_range), _kk_meta.csv (structured key/value parameters: smoothing, polyorder, lmin, metrics).
CLOSE	KK, Actions panel	Closes the window. The window is cached: reopening it resumes from the previous state.

5.7 Recommended workflow (4-step KK protocol)

The recommended operating protocol is:

- 1) fit_auto on $8^{\circ}+20^{\circ}$ → establishes d via geometric triangulation
- 2) Fix d
- 3) Iterative fit-KK loop (2-5 cycles typical):
 - a) Run a single FIT with d locked
 - b) Open KK Window, RECALCULATE
 - c) If $\text{RMS}/\sigma_{\text{KK}} > 3$ → APPLY KK→NODES (only the $\lambda < 420\text{-}470$ region)
 - d) If the 2550 node drifts too much → lock it by hand
 - e) Back to the main window, FIT again
- 4) FIT MULTI on $8^{\circ}+20^{\circ}$ with d fixed → produces the final n,k

Convergence criterion: $\text{RMS}(n - n_{\text{KK}})/\sigma_{\text{KK_local}}$ stabilized at the noise level ($\approx 2\text{--}3$) within a 5% variation between successive iterations.

6 Peak Analysis window

It opens by clicking **PEAK ANAL.** in the Main Window. Purpose: to analyze the reflectance maxima to estimate the thickness d in a model-independent way via Bragg/Snell, independently of the full fit.

6.1 Layout

Central plot (R vs λ with overlaid axes for n) + right panel with six groups of controls: Visibility, Global selection, Peak source, Load spectrum, Parameters, Actions, Calculation.

6.2 Hairline readout

Move the mouse over the plot and a vertical hairline follows the cursor; the box on the right lists, for the wavelength under the cursor, the R and n value of every visible curve. **Left-click to pin** the readout: the values stay on screen (marked **[PIN]**) even when you move the mouse away or off the window, until the next click. The box scrolls, and grows to show all rows when the panel has room.

6.3 Visibility matrix (4×4 toggle)

A 4×4 grid with a toggle for each combination of angle (8°/20°/40°/60°) and curve (**exp**, **fit**, **n**, **n_fit**). The **n_fit** column is the *fitted* $n(\lambda)$ curve (from per-angle ext_override), placed alongside **n** (n at the maxima) to compare the two methods. Clicking shows/hides the corresponding curve in the plot. Visual state: light green = visible, grey = hidden.

6.4 Global selection — ALL / NONE

Button	Where	What it does / when to use it
ALL	Peak Window, Global selection panel	Turns on all the Visibility toggles (shows all the curves).
NONE	Peak Window, Global selection panel	Turns off all the Visibility toggles (hides all the curves).

6.5 Peak source — FIT / EXP toggle

FIT button (yellow background if active, orange otherwise). When active the maxima are read from the fitted curve; when off, from the experimental data (**EXP**). The active source is shown in the plot title (' n from FIT/EXP maxima'), and switching it re-extracts the peaks and recomputes *both* the thickness d and $n(\lambda)$ from the maxima.

6.6 Load spectrum (per angle)

2×2 grid with a button per angle + status LED (green = data present, red = empty). Clicking a button opens a file dialog and loads the data for that angle, identical to the (folder) buttons in the main window.

6.7 Parameters

Two text fields:

- **Sample:** sample name used as prefix in SAVE CSV.
- **Y max (%):** upper limit of the R axis in the plot (default 65).

6.8 Actions

Button	Where	What it does / when to use it
UPDATE	Peak Window, Actions	Recomputes everything: finds the maxima on the currently visible curves, updates plot and annotations.
RESET	Peak Window, Actions	Resets Visibility (turns everything back on) and Peak source (returns to FIT).
SAVE CSV	Peak Window, Actions	Exports to CSV the peaks found for each angle (λ_{peak} , R_{peak} , order, ...).
CLOSE	Peak Window, Actions	Closes the window.

6.9 Calculation

Button	Where	What it does / when to use it
COMPUTE d (8/20)	Peak Window, Calculation	Bragg/Snell triangulation that solves for the thickness d from the 8° and 20° peaks (imposing the same maxima positions at both angles). Dispersion-free $\rightarrow$ robust against errors in $n(\lambda)$. It returns d only, not $n(\lambda)$.
CALC $n(\lambda)$	Peak Window, Calculation	Computes $n(\lambda)$ from the maxima at the <i>active</i> thickness. A two-radio selector next to the two d values picks the source: the computed (8/20) d , or a custom d you type in (e.g. the d_{eff} from the d-SCAN). Switching the radio does not overwrite the custom- d field. Output: $\lambda \leftrightarrow n$ table and a curve on the plot.

The $8^\circ/20^\circ$ triangulation produces d with very low sensitivity to the $n(\lambda)$ model: it is the methodological cornerstone of the PSi workflow. Use it as an independent anchor before FIT MULTI.

7 d-SCAN analysis window

Opened with the **d-SCAN** button (row 1). It is a **read-only** analysis over N curated sessions of the *same* sample, each fitted at a different thickness d : it does not re-fit, so its output is deterministic. Its job is to say which d the spectrum actually prefers, formalizing the expert's by-eye choice.

7.1 Input — the curated d-scan

Select several .psi_sess files of one sample that share the same loaded spectrum but were fitted at stepped d values (e.g. every 100 nm across the plausible range). Only the near-normal (8°) reflectance is used for d .

Do **not** use $40^\circ/60^\circ$ data to determine d : birefringence makes the effective index angle-dependent, which breaks the common-index assumption the thickness estimate relies on. The d-scan works on the 8° (near-normal) spectrum.

7.2 The ΔR d-discriminants (primary output)

Two *signed* metrics are computed against d , read directly off the curated fits:

- ΔC_{IR} — IR fringe-contrast difference (fit – exp) over the IR band (default 1000–2500 nm), where contrast = mean(fringe maxima) – mean(fringe minima). If d is too large, n comes out too small and the model under-contrasts the IR fringes $\rightarrow \Delta C_{IR} < 0$.
- ΔR — signed reflectivity-difference sum $\Sigma(R_{fit} - R_{exp})$ over an adjustable band (default 410–530 nm). If d is too small the model over-predicts R there $\rightarrow \Delta R > 0$.

Each metric changes sign as d passes the correct value, so its **zero-crossing brackets d** . In the figure the ΔC_{IR} curve uses the blue left axis and blue zero line, the ΔR curve the red right axis and red zero line; crossing markers can be toggled, and a green span highlights the bracket between the two crossings (its width **indicates** the bracket on d — honest, but relative to this indicator, not a rigorous uncertainty on d). The figures auto-save to _paper_figures/.

7.3 Band sliders

The ΔR band is a **RangeSlider** capped at 880 nm, i.e. just below the ~890 nm detector jump. The IR-contrast band is also slider-adjustable (experimental, so it can be removed later).

The band is sample-dependent — there is no single 'correct' band. Place it, by eye, on the region where the experimental and simulated curves visibly diverge in a d -dependent way (read it off the exp-vs-sim overlay), wide enough to give a clear slope, and always below the detector jump. A near-straight, steep $\Delta R(d)$ line crossing zero is the quality gate; a flat or wiggly line means that band is uninformative for that sample. The slider exists because this region *moves* with porosity — do not fix one band across samples.

7.4 Companion figures

- **$n(\lambda)$ overlay** — the fitted $n(\lambda)$ across the d-scan, to see how the recovered index tracks the assumed d . The *smoothest* curve(s) are highlighted: the physical $n(\lambda)$ follows a smooth normal (Cauchy) dispersion, so a d too small or too large distorts it into a flat shoulder or a sharp overshoot near the absorption edge. Curves are ranked by their RMS deviation from a Cauchy law over 420–1000 nm; as d is a **bracket**, the contiguous range of equally-smooth d is marked (one curve for a sharp minimum, its two edges for a flat one; a low-confidence note when even the best is irregular). A KK-free smoothness *indicator* — not a verdict: it cannot see anomalies outside 420–1000 nm, so a clearly rogue fit stays for you to reject.
- **exp vs sim reflectance** — the overlay used to place the ΔR band visually.
- **RMS-residual(d) / KK-residual window** — a complementary d -view: the **RMS reflectance residual** (%R) vs d with its parabolic minimum, and the KK residual on a twin axis. This RMS is an unweighted least-squares misfit — *not* a σ -normalized reduced χ^2 . The KK residual can be shown as a **magnitude**

($|\Delta n|$ RMS $\rightarrow$ a minimum) or, with a radio, **signed** ($\text{mean}(n_{\text{fit}} - n_{\text{KK}}) \rightarrow$ a zero-crossing, where n_{fit} passes from above to below n_{KK}), over a user-adjustable KK band (default 450–700 nm). Read it *alongside* ΔR : the RMS residual is more sensitive to the operator's fit choices than the physical ΔR , but on a well-curated scan the two agree — their convergence is what makes d solid.

The ΔR crossing is the lead single-angle d estimator: because it integrates a whole band rather than tracking individual peak positions, it is far steadier than the two-angle maxima. It is reliable for the robust quantities — d , and hence $n \cdot d$ and the overall shape of $n(\lambda)$, which the built-in KK/monotonicity/anchoring constraints keep self-consistent. The short-wavelength amplitude decomposition remains expert-in-the-loop. Treat the tool as a powerful companion to commercial instruments, not a replacement.

8 Operating workflows

Three typical workflows, in increasing order of rigour.

8.1 Quick mode — fit_auto only

For quick screening, process control, a first pass on a new sample: load at least 8° and 20°, press **FIT AUTO** and read *d*. Typical time: ~minutes. Reliable output: **d**. Approximate output: **(n, k)**.

8.2 Rigorous mode — fit_auto + KK cycles + fit_multi

For extracting optical constants intended for publication or optical design: apply the 4-step KK protocol described in the KK section. Typical time: ~10-20 minutes. Output: reliable **d** + KK-consistent **(n,k)** with a quantified RMS/ σ_{KK} metric.

8.3 Save → share → reload

To exchange the state with a colleague or for an operator-independence demonstration:

- 1) On the sending side:
 - FIT (or FIT MULTI/AUTO) for the angle(s) of interest
 - SAVE ALL → produces files *_all_8deg.par, *_all_8deg_data.csv, ... for each angle, + the manifest *_all.csv
 - Zip the folder, share it
- 2) On the receiving side:
 - Extract into the same folder structure
 - LOAD → select *_all.csv → automatically loads all the angles (R_exp only)
 - LOAD PAR → select one of the .par to also reload the parameters and nodes of the corresponding fit
 - FIT (or FIT MULTI) to verify reproducibility

For a complete internal backup (a single non-readable binary file) use **SAVE SESS / LOAD SESS** instead.

8.4 When to use which fit button

- **FIT**: optimization of a single angle. Fast, granular. Use it to refine an angle after manual slider changes.
- **FIT MULTI**: enforces consistency across angles (a single *n,k* for all). Use it when you have data at several angles and want a globally consistent solution. Slower.
- **FIT AUTO**: end-to-end orchestration with a *d* anchor via peak analysis. Starting point for a new sample.

9 Format of the files produced / read

9.1 File .par — single angle

Plain text, UTF-8 encoding. Sections:

```
# PSi Reflectometer parameter file
# Sample: <name> Angle: <ang> deg
# RMS residual (%R): <val>
# KK RMS(n-n_KK): <val> KK max|Dn|/n: <val> (if KK computed)
# Materials (symbolic names)
film_mat: PSi
substrate: Si
# Sample/instrument parameters
d: 1234.5678
scatt: 5.0000
inhom: 0.0000
offset: 1.0000
# IR correction
km: 1.0000
ks: 0.0000
kb: 0.0000
# Fringe contrast (envelope: amplitude, vertex, curvature)
fr_contrast: 1.0000
fr_ctr_vtx: 0.0000
fr_ctr_curv: 0.0000
# UV phase + angular spread (HOBL)
ps1: 0.0000
da: 0.0000
# W-nodes (movable base-node wavelengths)
w300: 300.0000
w335: 335.0000
w375: 375.0000
w400: 410.0000
w1100: 1100.0000
lmin: 450.0000
# Base nodes (14): wl_nm n k sigma_n sigma_k
240.00\t1.85\t0.10\t1.0e-02\t5.0e-04
270.00\t1.83\t0.08\t1.1e-02\t4.8e-04
... (14 rows in total)
```

The loader accepts an **incomplete** .par: a key it does not carry leaves that value unchanged, and fewer than 14 node rows load only the ones present (the σ columns are optional too). This is deliberate — you can hand-write a .par with just the parameters you care about (e.g. d and a few nodes) and load it to set only those: a versatility of the loader.

9.2 File _data.csv — single angle

CSV (delimiter ,), header in comments prefixed with # :

```
# PSi Reflectometer data file
# Sample: <name> Angle: <ang> deg
# d=<val>nm scatt=<val> inhom=<val>% offset=<val> RMS residual(%R)=<val>
# film_mat=PSi substrate=Si
# KK RMS=<val> KK max_rel=<val> (if KK computed)
# lambda_nm,R_exp_pct,R_fit_pct,n,k[,n_KK,k_KK][,sigma_n,sigma_k]
240.00,99.123456,99.012345,1.850000,0.100000
...
```

The KK columns appear only if the KK calculation has been done; the σ appear only if a fit with computed σ is available.

9.3 File `_all.csv` — multi-angle manifest

```
# PSi Reflectometer manifest
# Created: <YYYY-MM-DD>
sample: <name>
angles: 8,20,40
8: <name>_all_8deg_data.csv
20: <name>_all_20deg_data.csv
40: <name>_all_40deg_data.csv
```

The paths are **relative to the manifest folder**: if you move the files separately the link breaks. Move the whole folder.

9.4 File `.psi_sess` — complete session (binary)

Python pickle with a complete dump of the state (`data_angles`, `slider_vals`, `parameters`, `ext_override`, `kk_cache`, `ang_filenames`, ...). Not human-readable. Open only with LOAD SESS.

9.5 KK output files

Generated by the SAVE CURVES or SAVE ALL button of the KK window.

- **SAVE CURVES**: 1 CSV file with `wl`, `n_fit`, `n_KK_smooth`, `n_KK ± σ_KK`. Header with `smooth_window` and `lmin` in a comment.
- **SAVE ALL**: 3 files —
 - `_kk_grid.csv`: dense grid, columns: `wl_nm`, `n_fit`, `n_KK_raw`, `n_KK_smooth`, `k_input_Hilbert`, `sigma_KK_local`, `sigma_KK_propagated` (if available), `delta_n`.
 - `_kk_nodes.csv`: base+extended nodes with `wl_nm`, `n_fit`, `n_KK`, `delta_n`, `sigma_KK_local`, `delta_n_over_sigma`, `in_RMS_range`.
 - `_kk_meta.csv`: structured key/value parameters (`smooth_window`, `smooth_polyorder`, `lmin_RMS_nm`, `kk_rms`, `kk_max_rel`, `kk_sigma_kk`, `kk_snr`, `n_grid_points`, `n_nodes`, `sigma_prop_available`).

10 Physical constraints and priors

Three shape/validity choices are built into the program. Two are physical boundaries of the problem — the monotonicity boundary at 420 nm and the restriction of KK to near-normal incidence — and should not be moved. The third, $k \approx 0$ in the transparency region, is a prior you can switch off (the **K-BOUND** toggle).

10.1 Monotonicity of $n(\lambda)$ only at $\lambda \geq 420$ nm

PSi shows **anomalous dispersion** below 420 nm (UV region, near the interband transitions of Si). By construction the shape constraints on $n(\lambda)$ and $k(\lambda)$ (MONO n / MONO k toggles) apply only in the region $\lambda \geq 420$ nm. Do not extend the constraint below 420 nm even if the UV fit seems «improvable».

10.2 $k \approx 0$ in the transparency region (K-BOUND)

Above ~550 nm PSi is essentially transparent, so a finite k there is a calculation residue rather than physical information. The **K-BOUND** toggle (row 2, on by default) clamps k to ~0 in that region during a PSi fit; turn it off to let k float up to the graded physical bound. It affects only the free (PSi) fit — a known material keeps its tabulated k regardless.

10.3 KK valid only at 8°

The **KK** button of the Main Window opens the KK window **only if the active angle is 8°**. For larger angles the geometry departs from near-normal incidence and the assumptions of the KK relation derived from reflectance are no longer satisfied.

11 Glossary

- HOBL** — The «highly oblique» fit configurations: 40° and 60° angles. They add angular spread and birefringence (as Δn_{eff}); the model stays non-polarized throughout.
- AppState** — The Python object that centralizes all the program state (parameters, sliders, per-angle data, materials). Single, shared by reference.
- Base nodes** — The 14 λ nodes of the PCHIP model of n and k . Positions: 240, 270, 300, 335, 375, 430, 470, 520, 650, 900, 1100, 1583, 2067, 2550 nm. Some (300, 335, 375, 400, 1100) are modulated by the W sliders.
- Bragg/Snell triangulation** — Estimate of d via reflectance maxima at two angles (8°/20°) by imposing the same spectral position. Robust against the $n(\lambda)$ model.
- ext_override** — Tuple (ext_nodes, n_arr, k_arr, extra_dict) that stores the last fit. Used as the starting point by the next fit. Also stored per-angle in data_angles[ang].
- ext nodes** — Grid of ~40 nodes (14 base + 2 sub-nodes between each pair of base nodes) used internally by the fit. PCHIP-interpolated.
- Fade region** — Spectral region where the fringes attenuate due to absorption (increasing k). A constraint separate and independent from $n \cdot d$ for the determination of d .
- Hessian-diagonal σ** — Estimate of the uncertainties σ_n , σ_k at the nodes via numerical perturbation and computation of the diagonal of the χ^2 Hessian. Indicative, not rigorous.
- KK** — Kramers-Kronig. Integral relation between the real and imaginary parts of the complex dielectric function.
- kk_cache** — Triplet (wl_kk, n_kk_raw, k_kk) stored in state. n_kk_raw is the raw result of the Hilbert integral; smoothed n_{KK} is obtained by applying Savitzky-Golay.
- L-BFGS-B** — Quasi-Newton optimization algorithm with bound constraints used by the fit engine (scipy.optimize).
- Manifest** — Index file for multi-angle saving (see SAVE ALL). First line = marker; body = angle → file path map.
- PCHIP** — Piecewise Cubic Hermite Interpolating Polynomial. Monotone interpolation used to build continuous $n(\lambda)$, $k(\lambda)$ from the nodes.
- PSi** — Porous Silicon. Film with a columnar pore structure, n/k dependent on morphology.
- SAVE SESS** — Complete session save in binary format (pickle). For internal backup.
- Semi-infinite substrate** — Approximation of the substrate as infinitely thick; valid when the backside is scattering (unpolished PSi/Si).
- W-nodes** — The 5 base nodes with free λ position during PSi fits: W300, W335, W375, W400, W1100. For known films they stay fixed.